# System: iSCSI settings

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# System: iSCSI settings

## Concepts

### iSCSI terminology

Learn how the iSCSI terms apply to your storage array.

| Term | Description |
|---|---|
| CHAP | The Challenge Handshake Authentication Protocol (CHAP) method validates the identity of targets and initiators during the initial link. Authentication is based on a shared security key called a CHAP*secret*. |
| Controller | A controller consists of a board, firmware, and software. It controls the drives and implements the System Manager functions. |
| DHCP | Dynamic Host Configuration Protocol (DHCP) is a protocol used on Internet Protocol (IP) networks for dynamically distributing network configuration parameters, such as IP addresses. |
| IB | InfiniBand (IB) is a communications standard for data transmission between high-performance servers and storage systems. |
| ICMP PING response | Internet Control Message Protocol (ICMP) is a protocol used by operating systems of networked computers to send messages. ICMP messages determine whether a host is reachable and how long it takes to get packets to and from that host. |
| IQN | An iSCSI Qualified Name (IQN) identifier is a unique name for an iSCSI initiator or iSCSI target. |
| iSER | iSCSI Extensions for RDMA (iSER) is a protocol that extends the iSCSI protocol for operation over RDMA transports, such as InfiniBand or Ethernet. |
| iSNS | Internet Storage Name Service (iSNS) is a protocol that allows automated discovery, management, and configuration of iSCSI and Fibre Channel devices on TCP/IP networks. |
| MAC address | Media access control identifiers (MAC addresses) are used by Ethernet to distinguish between separate logical channels connecting two ports on the same physical transport network interface. |
| Management client | A management client is the computer where a browser is installed for accessing System Manager. |
| MTU | A Maximum Transmission Unit (MTU) is the largest size packet or frame that can be sent in a network. |

| Term | Description |
| --- | --- |
| RDMA | Remote Direct Memory Access (RDMA) is a technology that allows network computers to exchange data in main memory without involving the operating system of either computer. |
| Unnamed discovery session | When the option for unnamed discovery sessions is enabled, iSCSI initiators are not required to specify the target IQN to retrieve the controller's information. |

# How tos

## Configure iSCSI ports

If your controller includes an iSCSI host connection, you can configure the iSCSI port settings from the System page.

### Before you begin

- Your controller must include iSCSI ports; otherwise, the iSCSI settings are not available.
- You must know the network speed (the data transfer rate between the ports and the host).

> The iSCSI settings and functions only appear if your storage array supports iSCSI.

### Steps

1. Select **Settings › System**.
2. Under **iSCSI Settings**, select **Configure iSCSI Ports**.

   > The **Configure iSCSI Ports** option appears only if System Manager detects iSCSI ports on the controller.

3. Select the controller with the iSCSI ports you want to configure.
4. In the drop-down list, select the port you want to configure, and then click **Next**.
5. Select the configuration port settings, and then click **Next**.

   To see all port settings, click the **Show more port settings** link on the right of the dialog box.

**Field Details**

| Port Setting | Description |
| --- | --- |
| Enable IPv4 / Enable IPv6 | Select one or both options to enable support for IPv4 and IPv6 networks.<br><br>If you want to disable port access, deselect both check boxes. |
| TCP listening port (Available by clicking **Show more port settings**.) | If necessary, enter a new port number.<br><br>The listening port is the TCP port number that the controller uses to listen for iSCSI logins from host iSCSI initiators. The default listening port is 3260. You must enter 3260 or a value between 49152 and 65535. |
| MTU size (Available by clicking **Show more port settings**.) | If necessary, enter a new size in bytes for the Maximum Transmission Unit (MTU).<br><br>The default Maximum Transmission Unit (MTU) size is 1500 bytes per frame. You must enter a value between 1500 and 9000. |
| Enable ICMP PING responses | Select this option to enable the Internet Control Message Protocol (ICMP). The operating systems of networked computers use this protocol to send messages. These ICMP messages determine whether a host is reachable and how long it takes to get packets to and from that host. |

If you selected **Enable IPv4**, a dialog box opens for selecting IPv4 settings after you click **Next**. If you selected **Enable IPv6**, a dialog box opens for selecting IPv6 settings after you click **Next**. If you selected both options, the dialog box for IPv4 settings opens first, and then after you click **Next**, the dialog box for IPv6 settings opens.

6. Configure the IPv4 and/or IPv6 settings, either automatically or manually. To see all port settings, click the **Show more settings** link on the right of the dialog box.

**Field Details**

| Port setting | Description |
| --- | --- |
| Automatically obtain configuration | Select this option to obtain the configuration automatically. |
| Manually specify static configuration | Select this option, and then enter a static address in the fields. (If desired, you can cut and paste addresses into the fields.) For IPv4, include the network subnet mask and gateway. For IPv6, include the routable IP address and router IP address. |
| Enable VLAN support (Available by clicking **Show more settings**.) | Select this option to enable a VLAN and enter its ID. A VLAN is a logical network that behaves like it is physically separate from other physical and virtual local area networks (LANs) supported by the same switches, the same routers, or both. |
| Enable ethernet priority (Available by clicking **Show more settings**.) | Select this option to enable the parameter that determines the priority of accessing the network. Use the slider to select a priority between 1 (lowest) and 7 (highest).<br><br>In a shared local area network (LAN) environment, such as Ethernet, many stations might contend for access to the network. Access is on a first-come, first-served basis. Two stations might try to access the network at the same time, which causes both stations to back off and wait before trying again. This process is minimized for switched Ethernet, where only one station is connected to a switch port. |

7. Click **Finish**.

## Configure iSCSI authentication

For extra security in an iSCSI network, you can set authentication between controllers (targets) and hosts (initiators). System Manager uses the Challenge Handshake Authentication Protocol (CHAP) method, which validates the identity of targets and initiators during the initial link. Authentication is based on a shared security key called a CHAP *secret*.

### Before you begin

You can set the CHAP secret for the initiators (iSCSI hosts) either before or after you set the CHAP secret for the targets (controllers). Before you follow the instructions in this task, you should wait until the hosts have made an iSCSI connection first, and then set the CHAP secret on the individual hosts. After the connections are made, the IQN names of the hosts and their CHAP secrets are listed in the dialog box for iSCSI authentication (described in this task), and you do not need to manually enter them.

### About this task

You can select one of the following authentication methods:

- **One-way authentication** — Use this setting to allow the controller to authenticate the identity of the iSCSI

hosts (uni-directional authentication).

- **Two-way authentication** — Use this setting to allow both the controller and the iSCSI hosts to perform authentication (bi-directional authentication). This setting provides a second level of security by enabling the controller to authenticate the identity of the iSCSI hosts; and in turn, the iSCSI hosts to authenticate the identity of the controller.

> The iSCSI settings and functions only display on the Settings page if your storage array supports iSCSI.

**Steps**

1. Select **Settings › System**.
2. Under **iSCSI settings**, click **Configure Authentication**.

   The **Configure Authentication** dialog box appears, which shows the currently set method. It also shows if any hosts have CHAP secrets configured.

3. Select one of the following:
   - **No authentication** — If you do not want the controller to authenticate the identity of iSCSI hosts, select this option and click **Finish**. The dialog box closes, and you are done with configuration.
   - **One-way authentication** — To allow the controller to authenticate the identity of the iSCSI hosts, select this option and click **Next** to display the Configure Target CHAP dialog box.
   - **Two-way authentication** — To allow both the controller and the iSCSI hosts to perform authentication, select this option and click **Next** to display the Configure Target CHAP dialog box.
4. For one-way or two-way authentication, enter or confirm the CHAP secret for the controller (the target). The CHAP secret must be between 12 and 57 printable ASCII characters.

   > If the CHAP secret for the controller was configured previously, the characters in the field are masked. If necessary, you can replace the existing characters (new characters are not masked).

5. Do one of the following:
   - If you are configuring *one-way* authentication, click **Finish**. The dialog box closes, and you are done with configuration.
   - If you are configuring *two-way* authentication, click **Next** to display the Configure Initiator CHAP dialog box.
6. For two-way authentication, enter or confirm a CHAP secret for any of the iSCSI hosts (the initiators), which can be between 12 and 57 printable ASCII characters. If you do not want to configure two-way authentication for a particular host, leave the **Initiator CHAP Secret** field blank.

   > If the CHAP secret for a host was configured previously, the characters in the field are masked. If necessary, you can replace the existing characters (new characters are not masked).

7. Click **Finish**.

**Results**

Authentication occurs during the iSCSI login sequence between the controllers and iSCSI hosts, unless you specified no authentication.

## Enable iSCSI discovery settings

You can enable settings related to the discovery of storage devices in an iSCSI network. The Target Discovery Settings allow you to register the storage array's iSCSI information using the Internet Storage Name Service (iSNS) protocol, and also determine whether to allow unnamed discovery sessions.

### Before you begin

If the iSNS server uses a static IP address, that address must be available for iSNS registration. Both IPv4 and IPv6 are supported.

### About this task

You can enable the following settings related to iSCSI discovery:

- **Enable iSNS server to register a target** — When enabled, the storage array registers its iSCSI Qualified Name (IQN) and port information from the iSNS server. This setting allows iSNS discovery, so that an initiator can retrieve the IQN and port information from the iSNS server.
- **Enable unnamed discovery sessions** — When unnamed discovery sessions are enabled, the initiator (iSCSI host) does not need to provide the IQN of the target (controller) during the login sequence for a discovery-type connection. When disabled, the hosts do need to provide the IQN to establish a discovery-session to the controller. However, the target IQN is always required for a normal (I/O bearing) session. Disabling this setting can prevent unauthorized iSCSI hosts from connecting to the controller using only its IP address.

> The iSCSI settings and functions only display on the Settings page if your storage array supports iSCSI.

### Steps

1. Select **Settings › System**.
2. Under **iSCSI settings**, click **View/Edit Target Discovery Settings**.

   The **Target Discovery Settings** dialog box appears. Below the **Enable iSNS server**… field, the dialog box indicates if the controller is already registered.

3. To register the controller, select **Enable iSNS server to register my target**, and then select one of the following:
   - **Automatically obtain configuration from DHCP server** — Select this option if you want to configure the iSNS server using a Dynamic Host Configuration Protocol (DHCP) server. Be aware that if you use this option, all iSCSI ports on the controller must be configured to use DHCP as well. If necessary, update your controller iSCSI port settings to enable this option.

     > For the DHCP server to provide the iSNS server address, you must configure the DHCP server to use Option 43 — "Vendor Specific Information." This option needs to contain the iSNS server IPv4 address in data bytes 0xa-0xd (10-13).

   - **Manually specify static configuration** — Select this option if you want to enter a static IP address for the iSNS server. (If desired, you can cut and paste addresses into the fields.) In the field, enter either an IPv4 address or an IPv6 address. If you configured both, IPv4 is the default. Also enter a TCP listening port (use the default of 3205 or enter a value between 49152 and 65535).
4. To allow the storage array to participate in unnamed discovery sessions, select **Enable unnamed**

**discovery sessions**.

- When enabled, iSCSI initiators are not required to specify the target IQN to retrieve the controller's information.
- When disabled, discovery sessions are prevented unless the initiator provides the target IQN. Disabling unnamed discovery sessions provides added security.

5. Click **Save**.

#### Results

A progress bar appears as System Manager attempts to register the controller with the iSNS server. This process might take up to five minutes.

## View iSCSI statistics packages

You can view data about the iSCSI connections to your storage array.

#### About this task

System Manager shows these types of iSCSI statistics. All statistics are read-only and cannot be set.

- **Ethernet MAC statistics** — Provides statistics for the media access control (MAC). MAC also provides an addressing mechanism called the physical address or the MAC address. The MAC address is a unique address that is assigned to each network adapter. The MAC address helps deliver data packets to a destination within the subnetwork.
- **Ethernet TCP/IP statistics** — Provides statistics for the TCP/IP, which is the Transmission Control Protocol (TCP) and Internet Protocol (IP) for the iSCSI device. With TCP, applications on networked hosts can create connections to one another, over which they can exchange data in packets. The IP is a data-oriented protocol that communicates data across a packet-switched inter-network. The IPv4 statistics and the IPv6 statistics are shown separately.
- **Local Target/Initiator (Protocol) statistics** — Shows statistics for the iSCSI target, which provides block level access to its storage media, and shows the iSCSI statistics for the storage array when used as an initiator in asynchronous mirroring operations.
- **DCBX Operational States statistics** — Displays the operational states of the various Data Center Bridging Exchange (DCBX) features.
- **LLDP TLV statistics** — Displays the Link Layer Discovery Protocol (LLDP) Type Length Value (TLV) statistics.
- **DCBX TLV statistics** — Displays the information that identifies the storage array host ports in a Data Center Bridging (DCB) environment. This information is shared with network peers for identification and capability purposes.

You can view each of these statistics as raw statistics or as baseline statistics. Raw statistics are all of the statistics that have been gathered since the controllers were started. Baseline statistics are point-in-time statistics that have been gathered since you set the baseline time.

#### Steps

1. Select **Settings › System**.
2. Select **View iSCSI Statistics Packages**.
3. Click a tab to view the different sets of statistics.
4. **Optional:** To set the baseline, click **Set new baseline**.

Setting the baseline sets a new starting point for the collection of the statistics. The same baseline is used for all iSCSI statistics.

## View iSCSI sessions

You can view detailed information about the iSCSI connections to your storage array. iSCSI sessions can occur with hosts or remote storage arrays in an asynchronous mirror relationship.

**Steps**

1. Select **Settings › System**.
2. Select **View/End iSCSI Sessions**.

   A list of the current iSCSI sessions appears.

3. To see additional information about a specific iSCSI session, select a session, and then click **View Details**.

**Field Details**

| Item | Description |
|---|---|
| Session Identifier (SSID) | A hexadecimal string that identifies a session between an iSCSI initiator and an iSCSI target. The SSID is composed of the ISID and the TPGT. |
| Initiator Session ID (ISID) | The initiator part of the session identifier. The initiator specifies the ISID during login. |
| Target Portal Group | The iSCSI target. |
| Target Portal Group Tag (TPGT) | The target part of the session identifier. A 16-bit numerical identifier for an iSCSI target portal group. |
| Initiator iSCSI name | The worldwide unique name of the initiator. |
| Initiator iSCSI label | The user label set in System Manager. |
| Initiator iSCSI alias | A name that also can be associated with an iSCSI node. The alias allows an organization to associate a user-friendly string with the iSCSI name. However, the alias is not a substitute for the iSCSI name. The initiator iSCSI alias only can be set at the host, not in System Manager |
| Host | A server that sends input and output to the storage array. |
| Connection ID (CID) | A unique name for a connection within the session between the initiator and the target. The initiator generates this ID and presents it to the target during login requests. The connection ID is also presented during logouts that close connections. |
| Ethernet port identifier | The controller port associated with the connection. |
| Initiator IP address | The IP address of the initiator. |
| Negotiated login parameters | The parameters that are transacted during the login of the iSCSI session. |
| Authentication method | The technique to authenticate users who want access to the iSCSI network. Valid values are **CHAP** and **None**. |
| Header digest method | The technique to show possible header values for the iSCSI session. HeaderDigest and DataDigest can be either **None** or **CRC32C**. The default value for both is **None**. |
| Data digest method | The technique to show possible data values for the iSCSI session. HeaderDigest and DataDigest can be either **None** or **CRC32C**. The default value for both is **None**. |

| Item | Description |
| --- | --- |
| Maximum connections | The greatest number of connections allowed for the iSCSI session. The maximum number of connections can be 1 through 4. The default value is **1**. |
| Target alias | The label associated with the target. |
| Initiator alias | The label associated with the initiator. |
| Target IP address | The IP address of the target for the iSCSI session. DNS names are not supported. |
| Initial R2T | The initial ready to transfer status. The status can be either **Yes** or **No**. |
| Maximum burst length | The maximum SCSI payload in bytes for this iSCSI session. The maximum burst length can be from 512 to 262,144 (256 KB). The default value is **262,144 (256 KB)**. |
| First burst length | The SCSI payload in bytes for unsolicited data for this iSCSI session. The first burst length can be from 512 to 131,072 (128 KB). The default value is **65,536 (64 KB)**. |
| Default time to wait | The minimum number of seconds to wait before you attempt to make a connection after a connection termination or a connection reset. The default time to wait value can be from 0 to 3600. The default is **2**. |
| Default time to retain | The maximum number of seconds that connection is still possible following a connection termination or a connection reset. The default time to retain can be from 0 to 3600. The default value is **20**. |
| Maximum outstanding R2T | The maximum number of "ready to transfers" outstanding for this iSCSI session. The maximum outstanding ready to transfer value can be from 1 to 16. The default is **1**. |
| Error recovery level | The level of error recovery for this iSCSI session. The error recovery level value is always set to **0**. |
| Maximum receive data segment length | The maximum amount of data that either the initiator or the target can receive in any iSCSI payload data unit (PDU). |
| Target name | The official name of the target (not the alias). The target name with the *iqn* format. |
| Initiator name | The official name of the initiator (not the alias). The initiator name that uses either the *iqn* or *eui* format. |

4. **Optional:** To save the report to a file, click **Save**.

   The file is saved in the Downloads folder for your browser with the filename `iscsi-session-connections.txt`.

## End iSCSI session

You can end an iSCSI session that is no longer needed. iSCSI sessions can occur with hosts or remote storage arrays in an asynchronous mirror relationship.

### About this task

You might want to end an iSCSI session for these reasons:

- **Unauthorized access** — If an iSCSI initiator is logged on and should not have access, you can end the iSCSI session to force the iSCSI initiator off the storage array. The iSCSI initiator could have logged on because the None authentication method was available.
- **System downtime** — If you need to take down a storage array and you see that iSCSI initiators are still logged on, you can end the iSCSI sessions to get the iSCSI initiators off the storage array.

### Steps

1. Select **Settings › System**.
2. Select **View/End iSCSI Sessions**.

   A list of the current iSCSI sessions appears.

3. Select the session that you want to end.
4. Click **End Session**, and confirm that you want to perform the operation.

## Configure iSER over InfiniBand ports

If your controller includes an iSER over InfiniBand port, you can configure the network connection to the host.

### Before you begin

- Your controller must include an iSER over InfiniBand port; otherwise, the iSER over InfiniBand settings are not available in System Manager.
- You must know the IP address of the host connection.

### Steps

1. Select **Settings › System**
2. Under **iSER over InfiniBand settings**, select **Configure iSER over InfiniBand ports**.
3. Click the controller with the iSER over InfiniBand port you want to configure. Click **Next**.
4. In the drop-down list, select the HIC port you want to configure, and then enter the IP address of the host.
5. Click **Finish**.
6. Reset the iSER over InfiniBand port by clicking **Yes**.

## View iSER over InfiniBand statistics

If your storage array's controller includes an iSER over InfiniBand port, you can view data about the host connections.

### About this task

System Manager shows the following types of iSER over InfiniBand statistics. All statistics are read-only and cannot be set.

- **Local Target (Protocol) statistics** — Provides statistics for the iSER over InfiniBand target, which shows block-level access to its storage media.
- **iSER over InfiniBand Interface statistics** — Provides statistics for all iSER ports on the InfiniBand interface, which includes performance statistics and link error information associated with each switch port.

You can view each of these statistics as raw statistics or as baseline statistics. Raw statistics are all of the statistics that have been gathered since the controllers were started. Baseline statistics are point-in-time statistics that have been gathered since you set the baseline time.

### Steps

1. Select **Settings › System**.
2. Select **View iSER over InfiniBand Statistics**.
3. Click a tab to view the different sets of statistics.
4. **Optional:** To set the baseline, click **Set new baseline**.

   Setting the baseline sets a new starting point for the collection of the statistics. The same baseline is used for all iSER over InfiniBand statistics.

# FAQs

## What happens when I use an iSNS server for registration?

When Internet Storage Name Service (iSNS) server information is used, the hosts (initiators) can be configured to query the iSNS server to retrieve information from the target (controllers).

This registration provides the iSNS server with the controller's iSCSI Qualified Name (IQN) and port information, and allows for queries between the initiators (iSCSI hosts) and targets (controllers).

## Which registration methods are automatically supported for iSCSI?

The iSCSI implementation supports either the Internet Storage Name Service (iSNS) discovery method or the use of the Send Targets command.

The iSNS method allows for iSNS discovery between the initiators (iSCSI hosts) and targets (the controllers). You register the target controller to provide the iSNS server with the controller's iSCSI Qualified Name (IQN) and port information.

If you do not configure iSNS, the iSCSI host can send the Send Targets command during an iSCSI discovery session. In response, the controller returns the port information (for example, the Target IQN, port IP address,

listening port, and Target Port Group). This discovery method is not required if you use iSNS, because the host initiator can retrieve the target IPs from the iSNS server.

## How do I interpret iSER over InfiniBand statistics?

The View iSER over InfiniBand Statistics dialog box displays local target (protocol) statistics and iSER over InfiniBand (IB) interface statistics. All statistics are read-only, and cannot be set.

- **Local Target (Protocol) statistics** — Provides statistics for the iSER over InfiniBand target, which shows block-level access to its storage media.
- **iSER over InfiniBand Interface statistics** — Provides statistics for all iSER over InfiniBand ports on the InfiniBand interface, which includes performance statistics and link error information associated with each switch port.

You can view each of these statistics as raw statistics or as baseline statistics. Raw statistics are all of the statistics that have been gathered since the controllers were started. Baseline statistics are point-in-time statistics that have been gathered since you set the baseline time.

## What else do I need to do to configure or diagnose iSER over InfiniBand?

The following table lists the System Manager functions that you can use to configure and manage iSER over InfiniBand sessions.

> The iSER over InfiniBand settings are available only if your storage array's controller includes an iSER over InfiniBand host management port.

**Configure and diagnose iSER over InfiniBand**

| Action | Location |
|---|---|
| Configure iSER over InfiniBand ports | 1. Select **Hardware**.<br>2. Select **Show back of shelf**.<br>3. Select a controller.<br>4. Select **Configure iSER over InfiniBand ports**.<br>or<br>1. Select **Settings › System**.<br>2. Scroll down to **iSER over InfiniBand settings**, and then select **Configure iSER over InfiniBand Ports**. |
| View iSER over InfiniBand statistics | 1. Select **Settings › System**.<br>2. Scroll down to **iSER over InfiniBand settings**, and then select **View iSER over InfiniBand Statistics**. |

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